APPENDIX II

CENTRAL AFRICAN REPUBLIC

UPDATED DEBT SUSTAINABILITY ANALYSIS USING THE LOW-INCOME COUNTRY FRAMEWORK

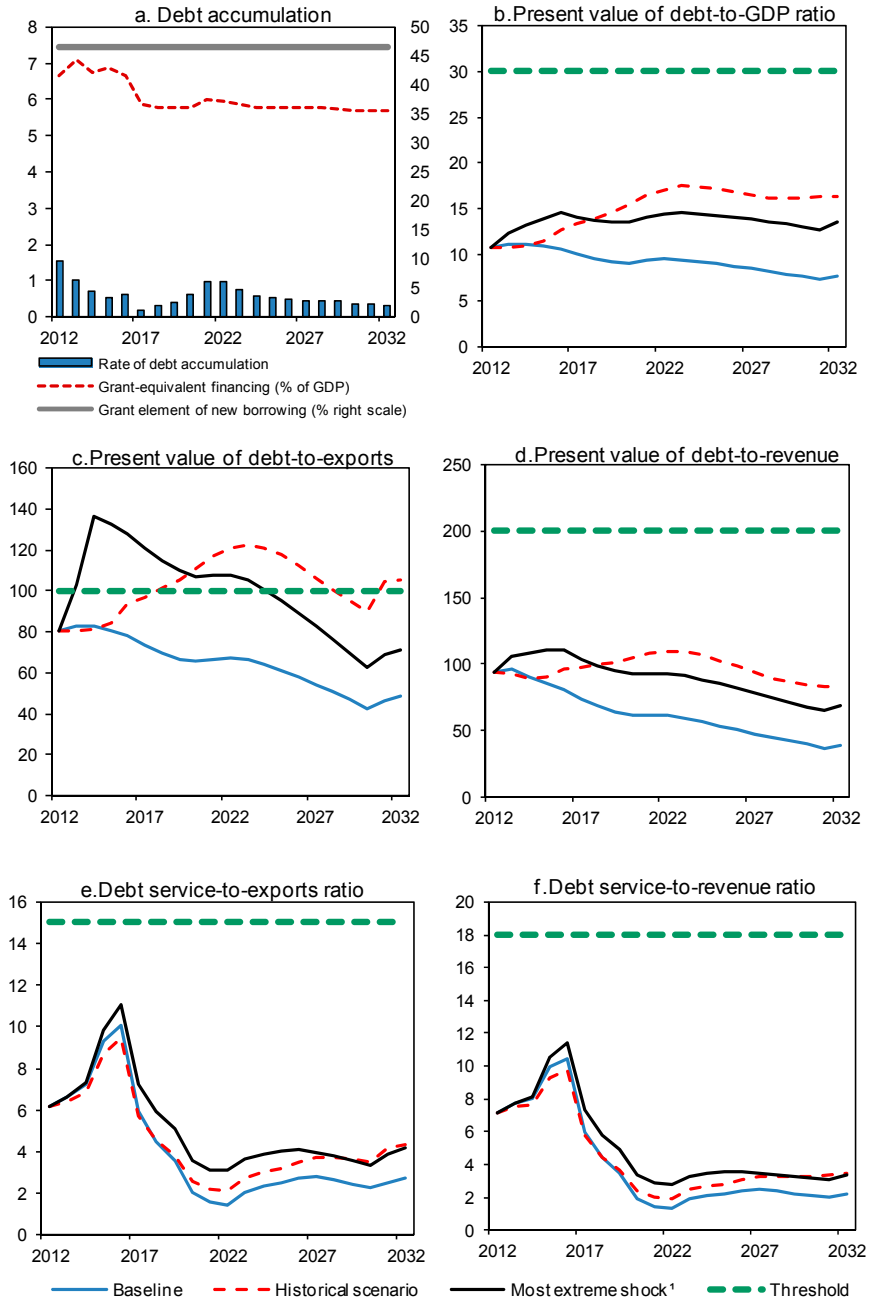
This appendix updates the debt sustainability analysis (DSA) originally presented in May 2010. Debt relief under the enhanced Heavily Indebted Poor Countries (HIPC) Initiative and the Multilateral Debt Relief Initiative (MDRI) at the completion point in June 2009 has alleviated Central African Republic's (C.A.R.) debt burden and the economy continues to face a moderate risk of debt distress.

C.A.R.'s total public debt, including domestic arrears increased markedly, reaching 42.2 percent of GDP in 2011. This is, well above the projected level of 27 percent in the 2010 DSA. The debt dynamic in 2010–11 is driven by arrears accumulation on both domestic and external debt and a steady rise in external borrowing from Non-Paris Club members to finance infrastructure projects and social programs. In 2010–11, the C.A.R. contracted concessional loans from India and China for US$52.3 million (2.4 percent of GDP). Accumulation of arrears on external debt amounted US$9.8 million (0.5 percent of GDP)—of which US$2 million is due to one Paris Club creditor (France) and US$0.7 million to the European Investment Bank, and the balance to Non-Paris Club creditors that have not yet signed bilateral agreements with C.A.R. Domestic debt accumulation amounted to 3.6 percent of GDP.

The macroeconomic assumptions in the near to medium term for the updated analysis are broadly in line with the previous DSA. Growth is expected to accelerate to around 5 percent per annum on average on the back of scaled-up infrastructures financed by concessional external loans and foreign direct investment inflows. The external current account is projected to remain steady as imports related to oil and infrastructure projects would offset rising exports. The fiscal stance is expected to improve on account of increased revenue mobilization and strong budget execution. From 2021 onwards, major mining projects (oil and uranium) would come on stream, boosting growth and government revenue.

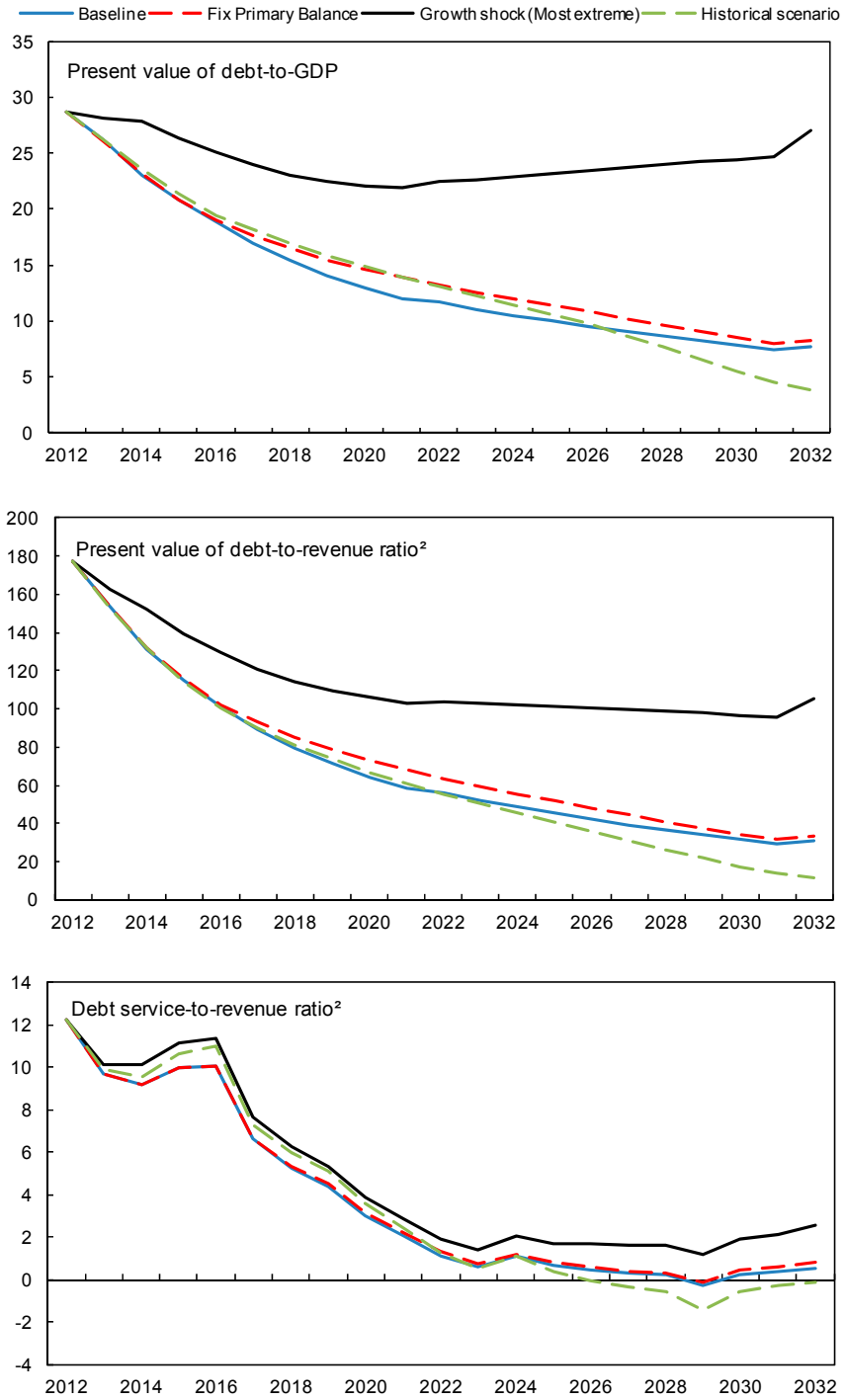
As in the full DSA, this update suggests that C.A.R. continues to face a "moderate risk of debt distress". Although somewhat higher compared to last year exercise, all baseline debt indicators remain below the thresholds with stable downward trends, but the debt position could still be vulnerable to the most extreme shock to terms of trade and exports, yielding the highest ratio by 2022. The C.A.R. is classified as having "lower debt vulnerabilities and lower capacity" under the Fund's debt limits framework and therefore should rely on concessional borrowing. Also, World Bank's country policy and institutional assessment rating for C.A.R. averaged 2.6 in the last three years, making C.A.R. a weak performer.

Figure II.1. Central African Republic: Indicators of Public and Publicly-Guaranteed External Debt Under Alternatives Scenarios, 2012–32 [1]



Sources: C.A.R. authorities; and IMF estimates and projections.
[1] The most extreme stress test is the test that yields the highest ratio in 2022. In figures b., d., e., and f. it corresponds to a terms-of-trade shock; in c. to an exports shock.

Figure II.2. Central African Republic: Indicators of Public Debt Under Alternative Scenarios, 2012–32 [1]



Sources: C.A.R. authorities; and IMF staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio in 2022.

[2] Revenues are defined inclusive of grants.

Table II.1.Central African Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009–32
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2009	2010	2011	Average [5]	Standard Deviation [5]	2012	2013	2014	2015	2016	2017	2012–17 Average	2022	2032	2018–32 Average
Public sector debt [1]	36.8	41.9	42.2			36.0	34.4	31.7	30.1	28.5	26.1		19.0	12.6	
o/w foreign-currency denominated	16.1	18.4	17.4			18.1	19.5	19.9	20.3	20.3	19.2		16.8	12.6	
Change in public sector debt	-43.5	5.1	0.3			-6.2	-1.6	-2.7	-1.6	-1.7	-2.4		-0.5	0.5	
Identified debt-creating flows	-6.5	1.0	1.4			-2.9	-1.0	-2.1	-1.1	-1.5	-2.3		0.0	0.2	
Primary deficit	-1.0	0.4	1.7	-0.6	3.7	0.5	0.6	0.1	0.8	0.5	-0.4	0.3	1.2	0.5	0.4
Revenue and grants	16.1	17.2	13.3			16.2	17.0	17.5	18.0	18.5	19.0		20.9	24.9	
of which: grants	5.3	5.5	2.5			4.6	5.4	5.3	5.3	5.3	5.3		5.3	5.3	
Primary (noninterest) expenditure	15.1	17.6	15.1			16.7	17.6	17.6	18.8	19.0	18.6		22.0	25.4	
Automatic debt dynamics	-4.8	0.4	-0.8			-3.1	-1.6	-2.2	-1.9	-2.0	-1.8		-1.4	0.0	
Contribution from interest rate/growth differential	-3.0	-0.8	-1.4			-1.6	-1.7	-2.4	-2.0	-2.0	-1.8		-1.4	0.0	
of which: contribution from average real interest rate	-1.6	0.3	-0.1			0.1	-0.3	-0.5	-0.4	-0.4	-0.3		-0.3	0.0	
of which: contribution from real GDP growth	-1.3	-1.1	-1.3			-1.7	-1.5	-1.9	-1.6	-1.6	-1.5		-1.1	0.0	
Contribution from real exchange rate depreciation	-1.8	1.2	0.6			-1.5	0.2	0.2	0.1	0.0	0.0		...	...	
Other identified debt-creating flows	-0.7	0.1	0.6			-0.3	0.0	0.0	0.0	0.0	0.0		0.3	-0.3	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.7	0.1	0.6			-0.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.3	-0.3	
Residual, including asset changes	-37.0	4.1	-1.1			-3.3	-0.6	-0.5	-0.5	-0.1	-0.1		-0.5	0.3	
Other Sustainability Indicators															
Present value (PV) of public sector debt	...	...	35.7			28.7	26.1	23.0	20.8	18.8	17.0		11.7	7.7	
o/w foreign-currency denominated	...	...	10.9			10.8	11.2	11.1	10.9	10.7	10.1		9.5	7.6	
o/w external	...	...	10.9			10.8	11.2	11.1	10.9	10.7	10.1		9.5	7.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need [2]	-0.6	1.3	2.3			2.5	2.2	1.7	2.6	2.3	0.8		1.4	0.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	267.9			177.7	153.8	131.1	115.2	101.6	89.6		56.2	30.8	
PV of public sector debt-to-revenue ratio (in percent)	…	…	331.0			248.2	225.1	188.2	163.6	142.6	124.6		75.4	39.1	
o/w external [3]	…	…	101.1			93.5	96.7	91.0	86.0	80.8	73.9		61.3	38.7	
Debt service-to-revenue and grants ratio (in percent) [4]	2.6	5.1	4.6			12.2	9.7	9.2	10.0	10.0	6.6		1.1	0.5	
Debt service-to-revenue ratio (in percent) [4]	3.9	7.5	5.7			17.1	14.2	13.2	14.2	14.1	9.2		1.5	0.7	
Primary deficit that stabilizes the debt-to-GDP ratio	42.4	-4.7	1.4			6.7	2.2	2.8	2.4	2.1	1.9		1.6	0.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.7	3.0	3.3	1.3	3.2	4.2	4.2	5.9	5.4	5.7	5.7	5.2	6.0	0.0	5.8
Average nominal interest rate on forex debt (in percent)	0.8	1.2	1.5	1.0	0.5	-0.1	0.0	0.0	0.1	0.2	0.2	0.1	0.5	0.7	0.5
Average real interest rate on domestic debt (in percent)	-0.5	2.0	-0.1	-0.5	2.1	1.3	0.5	-0.1	0.1	0.1	0.5	0.4	-0.3	-119.0	-14.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.4	7.8	3.3	-3.2	10.2	-9.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.6	2.1	1.9	3.0	1.9	1.2	2.1	2.6	2.2	2.2	2.0	2.1	2.3	0.0	2.1
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.2	-0.1	0.1	0.2	0.2	0.1	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	46.5	46.5	46.5	46.5	46.5	46.5	46.5	46.5	46.5	...

Sources: Country authorities; and IMF staff estimates and projections.

[1] General government, gross debt.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues excluding grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table II.2.Central African Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2012–32

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
Present value of debt-to-GDP ratio								
Baseline	29	26	23	21	19	17	12	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	29	26	24	21	19	18	13	4
A2. Primary balance is unchanged from 2012	29	26	23	21	19	18	13	8
A3. Permanently lower GDP growth [1]	29	26	23	21	20	18	15	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	29	28	28	26	25	24	23	27
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	29	28	26	24	22	20	14	9
B3. Combination of B1-B2 using one half standard deviation shocks	29	28	27	25	23	22	20	22
B4. One-time 30 percent real depreciation in 2013	29	30	26	23	20	18	12	7
B5. 10 percent of GDP increase in other debt-creating flows in 2013	29	32	28	26	24	22	16	11
Present value of debt-to-revenue ratio[2]								
Baseline	178	154	131	115	102	90	56	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	178	153	132	114	100	90	55	12
A2. Primary balance is unchanged from 2012	178	154	132	116	102	93	63	33
A3. Permanently lower GDP growth [1]	178	155	133	118	106	96	72	81
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	178	163	152	140	130	121	104	105
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	178	162	149	132	117	104	67	35
B3. Combination of B1-B2 using one half standard deviation shocks	178	161	147	133	123	113	92	87
B4. One-time 30 percent real depreciation in 2013	178	176	147	126	108	94	57	28
B5. 10 percent of GDP increase in other debt-creating flows in 2013	178	187	162	144	128	115	76	44
Debt service-to-revenue ratio[2]								
Baseline	12	10	9	10	10	7	1	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	10	10	11	11	7	1	0
A2. Primary balance is unchanged from 2012	12	10	9	10	10	7	1	1
A3. Permanently lower GDP growth [1]	12	10	9	10	10	7	1	2
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	12	10	10	11	11	8	2	3
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	12	10	9	10	10	7	1	1
B3. Combination of B1-B2 using one half standard deviation shocks	12	10	10	11	11	7	2	2
B4. One-time 30 percent real depreciation in 2013	12	11	12	13	13	9	2	2
B5. 10 percent of GDP increase in other debt-creating flows in 2013	12	10	10	11	10	7	1	1

Sources: Country authorities; and IMF staff estimates and projections.

[1] Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

[2] Revenues are defined inclusive of grants.

Table II.3a.: Central African Republic: External Debt Sustainability Framework, Baseline Scenario, 2009–32 [1]
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average [6]	Standard Deviation [6]	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) [1]	**16.1**	**18.4**	**17.4**			**18.1**	**19.5**	**19.9**	**20.3**	**20.3**	**19.2**		**16.8**	**12.6**	
o/w public and publicly guaranteed (PPG)	16.1	18.4	17.4			18.1	19.5	19.9	20.3	20.3	19.2		16.8	12.6	
Change in external debt	-40.8	2.3	-1.0			0.7	1.4	0.4	0.4	0.0	-1.1		0.0	0.5	
Identified net debt-creating flows	7.1	7.1	5.3			3.0	2.1	1.3	0.1	-1.1	-0.6		0.3	1.0	
Non-interest current account deficit	**8.7**	**10.0**	**8.5**	**5.4**	**3.6**	**6.8**	**5.9**	**5.3**	**5.2**	**4.6**	**4.1**		**5.1**	**6.2**	5.6
Deficit in balance of goods and services	-31.1	-35.2	-35.1			-37.0	-37.1	-36.5	-36.5	-36.1	-35.8		-37.5	-41.1	
Exports	9.5	10.6	11.9			13.4	13.5	13.5	13.6	13.7	13.8		14.2	15.6	
Imports	-21.5	-24.6	-23.3			-23.6	-23.6	-23.0	-22.9	-22.4	-22.0		-23.4	-25.5	
Net current transfers (negative = inflow)	-3.9	-4.0	-2.9	-3.9	1.1	-3.5	-4.2	-4.2	-4.2	-4.1	-4.1		-4.0	-3.5	-3.9
o/w official	-3.6	-3.7	-2.2			-3.1	-3.9	-3.9	-3.9	-3.9	-3.9		-3.9	-3.6	
Other current account flows (negative = net inflow)	43.7	49.2	46.5			47.3	47.2	46.0	45.8	44.8	44.0		46.7	50.8	
Net FDI (negative = inflow)	**-2.1**	**-3.1**	**-1.7**	**-2.6**	**1.4**	**-3.2**	**-3.1**	**-3.0**	**-4.1**	**-4.7**	**-3.7**		**-4.0**	**-5.3**	-4.3
Endogenous debt dynamics [2]	**0.5**	**0.2**	**-1.5**			**-0.7**	**-0.7**	**-1.1**	**-1.0**	**-1.0**	**-1.0**		**-0.9**	**0.1**	
Contribution from nominal interest rate	0.5	0.2	0.3			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.1	
Contribution from real GDP growth	-1.0	-0.5	-0.5			-0.7	-0.7	-1.1	-1.0	-1.1	-1.1		-0.9	0.0	
Contribution from price and exchange rate changes	1.0	0.4	-1.2			…	…	…	…	…	…		…	…	
Residual (3-4) [3]	**-48.0**	**-4.7**	**-6.3**			**-2.2**	**-0.7**	**-0.9**	**0.3**	**1.2**	**-0.5**		**-0.3**	**-0.5**	
o/w exceptional financing	-1.7	0.0	0.0			0.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt [4]	…	…	10.9			10.8	11.2	11.1	10.9	10.7	10.1		9.5	7.6	
In percent of exports	…	…	92.1			80.7	83.1	82.4	80.3	78.0	73.2		67.4	48.7	
PV of PPG external debt	**…**	**…**	**10.9**			**10.8**	**11.2**	**11.1**	**10.9**	**10.7**	**10.1**		**9.5**	**7.6**	
In percent of exports	**…**	**…**	**92.1**			**80.7**	**83.1**	**82.4**	**80.3**	**78.0**	**73.2**		**67.4**	**48.7**	
In percent of government revenues	**…**	**…**	**101.1**			**93.5**	**96.7**	**91.0**	**86.0**	**80.8**	**73.9**		**61.3**	**38.7**	
Debt service-to-exports ratio (in percent)	**-2.8**	**0.6**	**1.0**			**6.2**	**6.7**	**7.2**	**9.3**	**10.1**	**5.9**		**1.4**	**2.7**	
PPG debt service-to-exports ratio (in percent)	**-2.8**	**0.6**	**1.0**			**6.2**	**6.7**	**7.2**	**9.3**	**10.1**	**5.9**		**1.4**	**2.7**	
PPG debt service-to-revenue ratio (in percent)	**-2.5**	**0.6**	**1.1**			**7.1**	**7.8**	**8.0**	**10.0**	**10.5**	**6.0**		**1.3**	**2.2**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.1	0.2			0.1	0.1	0.1	0.1	0.0	0.0		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	49.6	7.7	9.5			6.1	4.5	5.0	4.8	4.5	5.2		5.1	5.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.7	3.0	3.3	1.3	3.2	4.2	4.2	5.9	5.4	5.7	5.7	5.2	6.0	0.0	5.8
GDP deflator in US dollar terms (change in percent)	-1.7	-2.7	7.0	7.8	7.7	4.1	1.2	1.6	1.4	2.2	2.0	2.1	2.3	0.0	2.1
Effective interest rate (percent) [5]	0.8	1.2	1.5	1.0	0.5	-0.1	0.0	0.0	0.1	0.2	0.2	0.1	0.5	0.7	0.5
Growth of exports of G&S (US dollar terms, in percent)	-13.5	11.4	23.7	5.7	13.0	22.7	6.2	7.4	7.7	8.9	8.4	10.2	9.3	0.0	9.1
Growth of imports of G&S (US dollar terms, in percent)	-8.2	14.3	4.6	10.1	10.9	10.1	5.3	4.9	6.5	5.7	5.8	6.4	11.8	0.0	9.2
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	46.5	46.5	46.5	46.5	46.5	46.5	46.5	46.5	46.5	46.5
Government revenues (excluding grants, in percent of GDP)	10.8	11.6	10.8			11.6	11.6	12.2	12.7	13.2	13.6		15.5	19.7	16.9
Aid flows (in Billions of US dollars) [7]	0.1	0.1	0.1			0.2	0.2	0.2	0.2	0.2	0.2		0.3	0.6	
o/w Grants	0.1	0.1	0.1			0.1	0.1	0.1	0.2	0.2	0.2		0.3	0.6	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.1	0.1	0.1	0.1	0.0		0.1	0.1	
Grant-equivalent financing (in percent of GDP) [8]	…	…	…			6.7	7.1	6.8	6.9	6.7	5.8		6.0	5.7	5.8
Grant-equivalent financing (in percent of external financing) [8]	…	…	…			73.7	78.4	80.1	79.6	81.1	90.8		88.7	92.7	91.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	2.0	2.0	2.2			2.4	2.5	2.7	2.9	3.1	3.4		5.0	10.8	
Nominal dollar GDP growth	-0.1	0.2	10.5			8.5	5.5	7.6	6.9	8.0	7.8	7.4	8.4	0.0	8.1
PV of PPG external debt (in Billions of US dollars)			0.2			0.3	0.3	0.3	0.3	0.3	0.3		0.5	0.8	
(PVt-PVt-1)/GDPt-1 (in percent)						1.5	1.0	0.7	0.5	0.6	0.2	0.8	1.0	0.3	0.5
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	…	…	11.0			10.9	11.3	11.2	11.0	10.7	10.1		9.6	7.6	
PV of PPG external debt (in percent of exports + remittances)	…	…	95.4			83.8	86.3	85.5	83.3	80.8	75.7		69.0	49.3	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	1.0			6.4	6.9	7.5	9.6	10.5	6.1		1.5	2.8	

Sources: Country authorities; and IMF staff estimates and projections.

[1] Includes both public and private sector external debt.

[2] Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

[4] Assumes that PV of private sector debt is equivalent to its face value.

[5] Current-year interest payments divided by previous period debt stock.

[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

[7] Defined as grants, concessional loans, and debt relief.

[8] Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table II.3b.Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2012–32
(Percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
Present value of debt-to GDP ratio								
Baseline	11	11	11	11	11	10	**10**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 [1]	11	11	11	11	13	13	**17**	16
A2. New public sector loans on less favorable terms in 2012-2032 [2]	11	12	13	14	15	14	**15**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	11	12	13	12	12	12	**11**	9
B2. Export value growth at historical average minus one standard deviation in 2013-2014 [3]	11	12	14	14	13	13	**11**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	11	11	11	11	11	10	**10**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 [4]	11	13	15	15	14	13	**12**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	11	13	16	16	15	14	**13**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	11	16	16	15	15	14	**14**	11
Present value of debt-to-exports ratio								
Baseline	81	83	82	80	78	73	**67**	49
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 [1]	81	80	81	85	94	97	**121**	105
A2. New public sector loans on less favorable terms in 2012-2032 [2]	81	91	98	103	106	103	**103**	86
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	81	83	82	80	78	73	**67**	48
B2. Export value growth at historical average minus one standard deviation in 2013-2014 [3]	81	103	136	132	128	121	**107**	71
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	81	83	82	80	78	73	**67**	48
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 [4]	81	97	110	107	103	97	**86**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	81	103	130	126	122	115	**101**	65
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	81	83	82	80	78	73	**67**	48
Present value of debt-to-revenue ratio								
Baseline	94	97	91	86	81	74	**61**	39
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 [1]	94	93	89	91	97	98	**110**	83
A2. New public sector loans on less favorable terms in 2012-2032 [2]	94	106	108	110	110	104	**93**	69
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	94	102	104	98	92	84	**70**	44
B2. Export value growth at historical average minus one standard deviation in 2013-2014 [3]	94	105	114	107	100	92	**74**	43
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	94	97	93	88	83	76	**63**	40
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 [4]	94	113	122	114	107	98	**78**	44
B5. Combination of B1-B4 using one-half standard deviation shocks	94	114	131	123	115	106	**84**	47
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	94	137	129	122	114	105	**87**	55

(Contn.d)

Table II.3b.Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2012-2032 *(conclued)*
(Percent)

Debt service-to-exports ratio								
Baseline	6	7	7	9	10	6	**1**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 [1]	6	6	7	9	9	6	**2**	4
A2. New public sector loans on less favorable terms in 2012-2032 [2]	6	7	7	10	11	7	**3**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	6	7	7	9	10	6	**1**	3
B2. Export value growth at historical average minus one standard deviation in 2013-2014 [3]	6	8	10	13	14	8	**2**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	6	7	7	9	10	6	**1**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 [4]	6	7	7	10	11	6	**2**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	6	7	9	11	12	7	**2**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	6	7	7	9	10	6	**1**	3
Debt service-to-revenue ratio								
Baseline	7	8	8	10	10	6	**1**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 [1]	7	7	8	9	10	6	**2**	3
A2. New public sector loans on less favorable terms in 2012-2032 [2]	7	8	8	11	11	7	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	7	8	9	11	12	7	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2013-2014 [3]	7	8	8	10	11	6	**1**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	7	8	8	10	11	6	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 [4]	7	8	8	10	11	6	**2**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	7	8	9	11	12	7	**2**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 [5]	7	11	11	14	15	8	**2**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) [6]	44	44	44	44	44	44	**44**	44

Sources: Country authorities; and IMF staff estimates and projections.

[1] Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
[2] Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
[3] Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
[4] Includes official and private transfers and FDI.
[5] Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
[6] Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.